

FFGFT — Information and Black Holes

What Is Fundamental Information? Ontology vs. Epistemology

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Abstract

This document clarifies what *information* means in FFGFT in the context of black holes. It shows that the Hawking information paradox rests on a category confusion: the classical formulation treats information as an epistemic concept. FFGFT distinguishes **fundamental information**: the topological structure of the winding numbers on T^4 — comparable to a topological charge, not to a quantity of bits. This topological charge is **topologically preserved**, independent of any observer.

1 The Hawking Information Paradox and Its Presuppositions

Stephen Hawking showed in 1974 that black holes emit thermal radiation, carrying no information about the specific state of the infallen matter. If the black hole evaporates completely, the original quantum information would be destroyed — contradicting the unitarity of quantum mechanics.

The paradox presupposes three things, not all of which hold in FFGFT:

- Information is the complete description of the quantum state of all infallen particles.
- Time is continuous and one-dimensionally directed.
- The classical singularity is physically real — matter is destroyed there.

Why does the paradox arise at all?

Hawking presupposes that information is *epistemic* and that time is continuously directed. Neither is fundamental in FFGFT: information is a topological charge; time is compactified on T^4 .

2 What Information in FFGFT Is Not

2.1 Not our description of the black hole

Information in the FFGFT sense is **not** our knowledge of the black hole. The Hawking paradox is an **epistemic** problem: can an observer reconstruct the original quantum state? FFGFT answers a different question: does the fundamental topological structure still exist?

Information in the FFGFT sense is not an epistemic category but an **ontological** one.

Observable consequence: If the topological information were truly destroyed, the Hawking radiation would be exactly black-body. The modified spectrum is the falsifiable difference.

2.2 Not the sum of all detail states

Information is also not the sum of all detail states. These can be epistemically lost through thermalisation — thermodynamic irreversibility, not ontological information loss.

3 What Information in FFGFT Is: Fundamental Information

3.1 Topological invariants as fundamental information

In FFGFT, particles are winding configurations on T4 with quantum numbers ($n_\theta, n_\phi, n_3, n_4$) — **topological invariants**.

Definition of fundamental information

Fundamental information is the **topological charge**: the winding numbers on T4. They define *what kind* of particles exist. Not a *quantity of bits* (Shannon), but a topological charge — like a winding number in string theory.

No annihilation operator acts on topological invariants. No process can reduce winding numbers to zero without a measurable discontinuity. Singularities are therefore structurally excluded.

3.2 The foundational relation as structure preservation

The fundamental relation $T \sim m = 1$ links the T-field and mass. In the limit of extreme compression:

$$T \rightarrow L_0/c = x_i \cdot t_P > 0$$

The T-field never reaches zero — the minimum is L_0/c . The winding structure is preserved.

3.3 Algebraic protection against singularities (Dok. 078)

From $T \sim m = 1$ it follows that $T = 1/m$. If $m \rightarrow \text{infinity}$, then $T \rightarrow 0$. But reaching $T = 0$ exactly would require $m = \text{infinity}$ exactly — algebraically impossible.

Algebraic protection — Dok. 078

$T \sim m = 1$ acts as algebraic protection:

- $m \rightarrow \text{infinity}$ implies $T \rightarrow 0$ — only **asymptotically**, never exactly.
- The singularity is not a forbidden state but an **unreachable limit**.

3.4 The interior operator: what happens at L_0

T-field at the minimum: $T_{\min} = x_i \cdot t_P > 0$ — finite, positive, non-singular.

Maximal winding density: one winding quantum per L_0^4 . Densest stable configuration ($N = 0$). Below this the state space ends.

Topological bounce condition: no stable configurations below $N = 0$ — a state-space boundary, not a force mechanism.

Time in the interior: $T \sim \rightarrow L_0/c$. Time does not freeze but reaches its smallest value. The modified Hawking spectrum arises at this interface.

Interior of a black hole in FFGFT

The interior is an extreme state:

- $T_{\min} = \xi \cdot t_P > 0$ — finite, non-singular
- Maximal winding density $1/L_0^4$
- State-space boundary at $N = 0$
- Unitary transition operator (details open)

4 Black Holes as Inversion — Not as Destruction

A black hole is not a destruction state but an **inversion**: winding configurations maximally compressed, T-field at the minimum L_0/c . What fell in as an electron remains topologically classifiable as an electron.

Irreversibility and information preservation operate on different levels:

- **Thermodynamic:** directed, irreversible, entropy increases.
- **Topological:** winding numbers preserved.

The Hawking paradox arises from confusing these levels.

5 The Radiation Mechanism: Node Tunnelling and Modified Spectrum

The preceding sections clarified what happens to the *information*. This section clarifies the *mechanism* of the radiation itself — and derives the FFGFT-specific spectrum modification from first principles.

5.1 Temperature and mechanism

The Hawking temperature remains **unchanged** in FFGFT (Dok. 201, Sec. 7.4):

$$T_H = \hbar \kappa / (2\pi k_B) = \hbar c^3 / (8\pi G M k_B), \quad \kappa = c^4 / (4GM)$$

The surface gravity κ is geometrically fixed. What changes is the **emission mechanism**: instead of particle-antiparticle pair production, FFGFT emits **incoherent node excitations** that **tunnel** through the phase barrier at the horizon (Dok. 201, Sec. 7.4; Dok. 049). This is the FFGFT translation of the Parikh-Wilczek tunnelling picture: the quanta are winding excitations of the T-field, not modes of a continuous field.

Numerically ($M = M_{\text{Sun}}$): T_H about 6.17×10^{-8} K; r_s about 2.95 km. Since T_H is proportional to $1/M$, evaporation accelerates as the mass decreases.

5.2 Entropy as node count

The Bekenstein-Hawking entropy is the **configuration entropy of the nodes**:

$$S = A/(4 \cdot l_P^2) = N_{\text{node}} \cdot \ln 2, \quad N_{\text{node}} \propto A/x_i^2$$

Each node carries one bit ($\ln 2$). For $M = M_{\text{Sun}}$: S/k_B about 1.05×10^{77} , so N_{node} about 1.5×10^{77} . This reproduces the area law exactly — no free adjustment.

5.3 Derivation of the spectrum modification from lattice dispersion

The central FFGFT prediction follows not from an ansatz but from the **minimal length** $L_0 = x_i l_P$. A minimal length implies a discrete lattice structure with a modified dispersion relation (analogous to phonon dispersion, Debye theory):

$$E(k) = (2 \hbar c / L_0) \sin(k L_0 / 2) = \hbar c k [1 - (k L_0)^2 / 24 + \dots]$$

The leading correction:

$$dE/E = -(k L_0)^2 / 24 = -(1/24) (E/E_{\text{max}})^2, \quad E_{\text{max}} = \hbar c / L_0 = E_P / x_i$$

The cutoff $E_{\text{max}} = E_P / x_i$ about $7500 E_P$ about 9.2×10^{22} GeV lies above the Planck energy. In units of the Hawking temperature ($E = x k_B T_H$):

$$dE/E = -(x^2 / 24) (k_B T_H / E_{\text{max}})^2$$

5.4 Scale and falsifiability — an honest assessment

This derivation has the correct limiting behaviour, but its practical consequence must be **stated honestly**: for real black holes the modification is unmeasurably small. For $M = M_{\text{Sun}}$:

$$k_B T_H / E_{\text{max}} \text{ about } 5.8 \times 10^{-44} \Rightarrow dE/E \text{ (at } x=10) \text{ about } -1.4 \times 10^{-86}$$

The modification becomes significant only when $k_B T_H$ approaches E_{max} — that is, at $M_{\text{crit}} = c^2 L_0 / (8 \pi G)$ about 1.15×10^{-13} kg, the **remnant scale**. The modification arises exactly at the L_0 interface.

Correction of a naive ansatz

A naive modulation ansatz $\exp(-x_i x^2 / D_f)$ yields artificially large deviations (about 0.4%) that physically **do not exist**. The correct deviation scales as $(k_B T_H / E_{\text{max}})^2$ and is about 10^{-86} . A theory claiming 0.4% would already be falsified. The correct prediction: the spectrum is modified only at the evaporation endpoint.

5.5 Evaporation endpoint: remnant rather than destruction

Near M_{crit} the system reaches the state-space boundary $N = 0$: no denser stable configurations. The result is a **stable remnant** (Dok. 201, Sec. 7.5):

- finite size about L_0 ,
- finite temperature,
- topological charge preserved in the remnant node state.

The fundamental information remains encoded **on this side**. There is no transfer to an "other side" and no destruction. The thermodynamic time direction remains irreversible; topological preservation is unaffected.

6 The Compactified Time Dimension

In FFGFT the time dimension is **compactified** on T^4 — the winding number n_4 is a topological invariant.

Compactification means **periodicity of field configurations** in the time direction — not reversibility of thermodynamic time. A macroscopic process remains irreversible. The T^4 topology is time-direction-neutral; the thermodynamic arrow is emergent.

The experienced arrow of time is a thermodynamic phenomenon (entropy increase), not a geometric primitive. n_4 remains topologically preserved, independent of the observer time direction.

7 What FFGFT Concretely Claims — and What It Does Not

- **No classical singularity:** $L_0 = \xi \cdot l_P$ as absolute lower density bound.
- **Topological fundamental information preserved:** winding numbers as topological invariants, algebraically protected by $T \sim m = 1$.
- **Modified Hawking spectrum:** derived from lattice dispersion ($dE/E = -(E/E_{\max})^{2/24}$). Unmeasurable for real black holes; significant only at the remnant scale.
- **Gravitational-wave echoes:** after mergers as a further test.

Not claimed: complete state reconstruction; a fully specified transition operator near $N = 0$; suspension of thermodynamic irreversibility.

Precise FFGFT statement

FFGFT claims: the **topological fundamental charge** is **topologically preserved**. It is not reconstructable by an external observer (access problem) but exists as a geometric property of the T^4 field. If destroyed, the Hawking spectrum would be exactly thermal — the modified spectrum is the measurable difference.

8 Connection to the ξ Recursion

The ξ ladder defines which winding configurations on T^4 are stable. Black holes represent configurations at the lower end ($N = 0$) — within the ξ structure, not outside it.

Python-Verified Key Numbers

Quantity	Value
ξ	1.333×10^{-4}
$L_0 = \xi \cdot l_P$	$2.155 \times 10^{-39} \text{ m}$
$T_{\sim \min} = \xi \cdot t_P$	$7.19 \times 10^{-48} \text{ s}$
$T_H (M_{\text{Sun}})$	$6.17 \times 10^{-8} \text{ K}$
$S/k_B (M_{\text{Sun}})$	1.05×10^{77}
$N_{\text{node}} (M_{\text{Sun}})$	1.5×10^{77}
$E_{\text{max}} = \hbar \cdot c / L_0$	$9.2 \times 10^{22} \text{ GeV} (= E_P / \xi)$
$M_{\text{crit}} (\text{remnant})$	$1.15 \times 10^{-13} \text{ kg}$
$dE/E (M_{\text{Sun}}, x=10)$	$-1.4 \times 10^{-86} \text{ (unmeasurable)}$

Summary

Question	FFGFT Answer
Why does the paradox arise?	Hawking presupposes information as epistemic and time as continuous
What is fundamental information?	Topological charge: winding numbers on T^4 — not bits
Is information lost?	Topological charge: no. Detail state: access problem
Radiation mechanism	Node tunnelling through phase barrier (Parikh-Wilczek); T_H unchanged
Spectrum modification	From lattice dispersion: $dE/E = -(E/E_{\text{max}})^2/24$; about 10^{-86} unmeasurable
Algebraic protection (Dok. 078)	large m implies $T_{\sim} \rightarrow 0$ only asymptotically — singularity unreachable
Time compactification	Periodicity of field configurations, not reversal of time
Falsifiable tests	Hawking spectrum (not exactly thermal); gravitational-wave echoes
What remains open?	Transition operator at L_0 — unitary, details open